Customer Segmentation with Clustering

Mini-Project 6.3 | CAM DS101 – Week 6
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Introduction **1**
Dataset Description **1**
Methodology **2**
Feature Engineering 2
EDA and Preprocessing 3
Optimal Number of Clusters 4
Results **5**
Dimensionality Reduction 6
Conclusion and Recommendations **8**
References **8**

Introduction

This report presents the findings of a customer segmentation analysis performed on a real-world e-commerce dataset. Customer segmentation is the process of grouping customers with shared characteristics and is a critical strategic tool for retail businesses. It enables targeted marketing, improved customer retention, price optimisation, and smarter resource allocation. The objective of this project was to apply unsupervised machine learning clustering techniques to identify meaningful customer segments and provide the business with actionable marketing recommendations.

Dataset Description

The dataset contains 951,669 transactional records spanning April 2012 to October 2016, representing 68,300 unique customers across 47 countries on five continents (Europe, North America, Oceania, Asia, and Africa). Each row represents one ordered product and includes 20 features covering order details (date, revenue, unit cost, discount), customer demographics

(birthdate, country, loyalty group), and order metadata. Following the removal of 21 duplicate rows, 951,648 clean records remained. Missing values were found only in City (135), Postal_Code (3,716) and State_Province (117,192). Although these features were not used in the model, Postal_Code and State_Province missing values were filled with ‘unknown’ tags. While City values were filled with the mode of any matches to the Postal_Code or State_Province for that row. After feature engineering and aggregation, the dataset comprised 68,300 unique customer records.

Methodology

The analysis followed a structured pipeline. Starting with data cleaning, feature engineering, EDA. Before moving on to preprocessing, optimal cluster selection, K-Means clustering, and dimensionality reduction for visualisation.

Feature Engineering

Five features were derived from the transactional data:

• Frequency: Count of orders per customer — a marker for engagement and loyalty.

• Recency: Days between the customer's most recent delivery and today. The data for recency was also inverted so higher score = more recent. This was to make all features have the same scale and prevent any issues this might cause with clustering.

• CLV (Customer Lifetime Value): Sum of all revenue generated by the customer — a direct measure of business value.

• Avg_Unit_Cost: Mean unit cost per order — an indicator of purchasing behaviour and product tier preference.

• Age: Customer age in years, calculated from birth date — a demographic feature.

EDA and Preprocessing

EDA revealed right skew and outliers in Frequency, CLV and Avg_Unit_Cost. There was a large present of extreme outliers in the data which were capped using the 3×IQR method. This was chosen to only remove the most extreme outliers. Recency had multimodal peaks in the feature distribution, which may be linked to previous marketing campaigns.Age appeared normally distributed with no outliers. See Figure 1 and Figure 2 for the EDA plots.

All five features were then standardised using StandardScaler to ensure equal contribution to the distance-based K-Means algorithm.

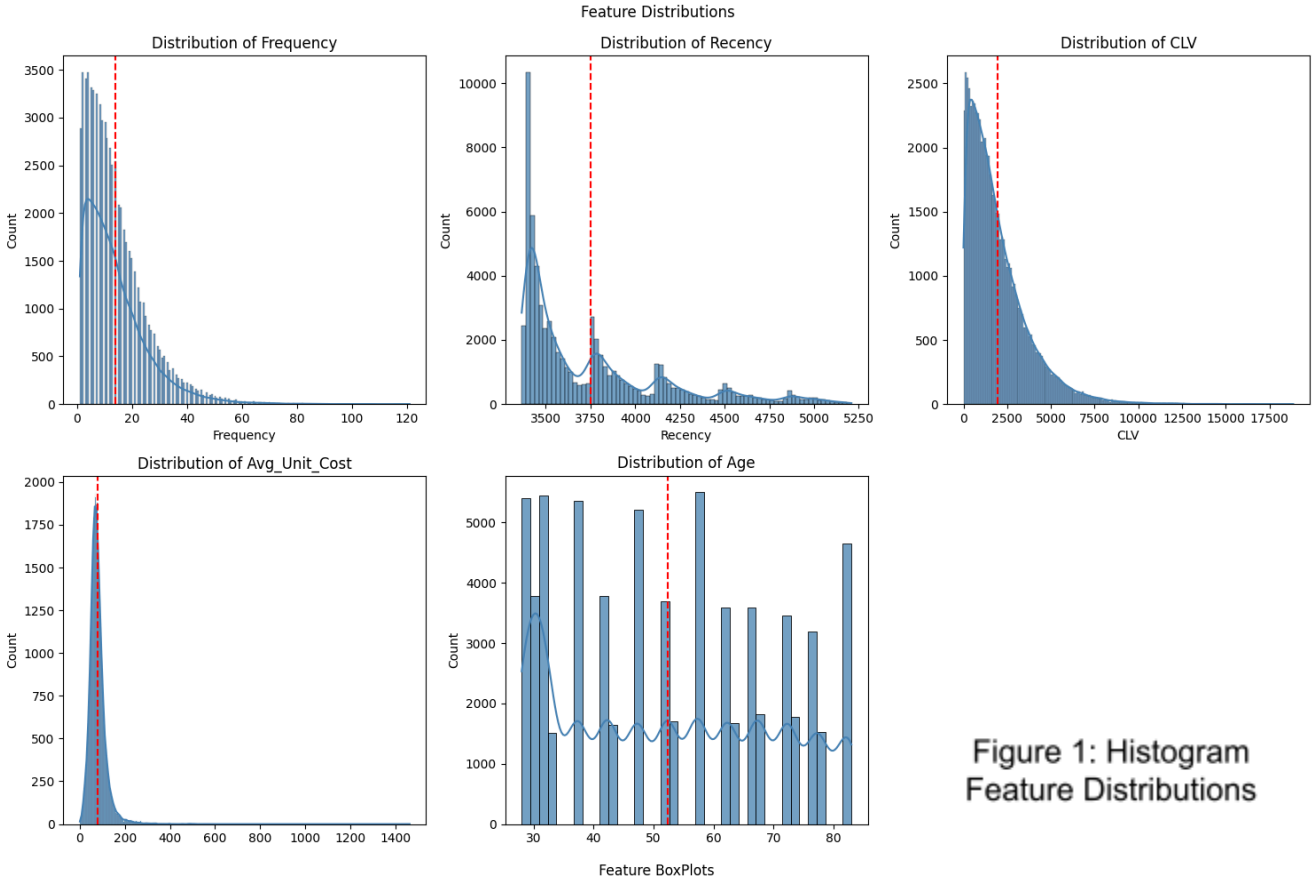

Figure 1: Histogram Feature Distributions

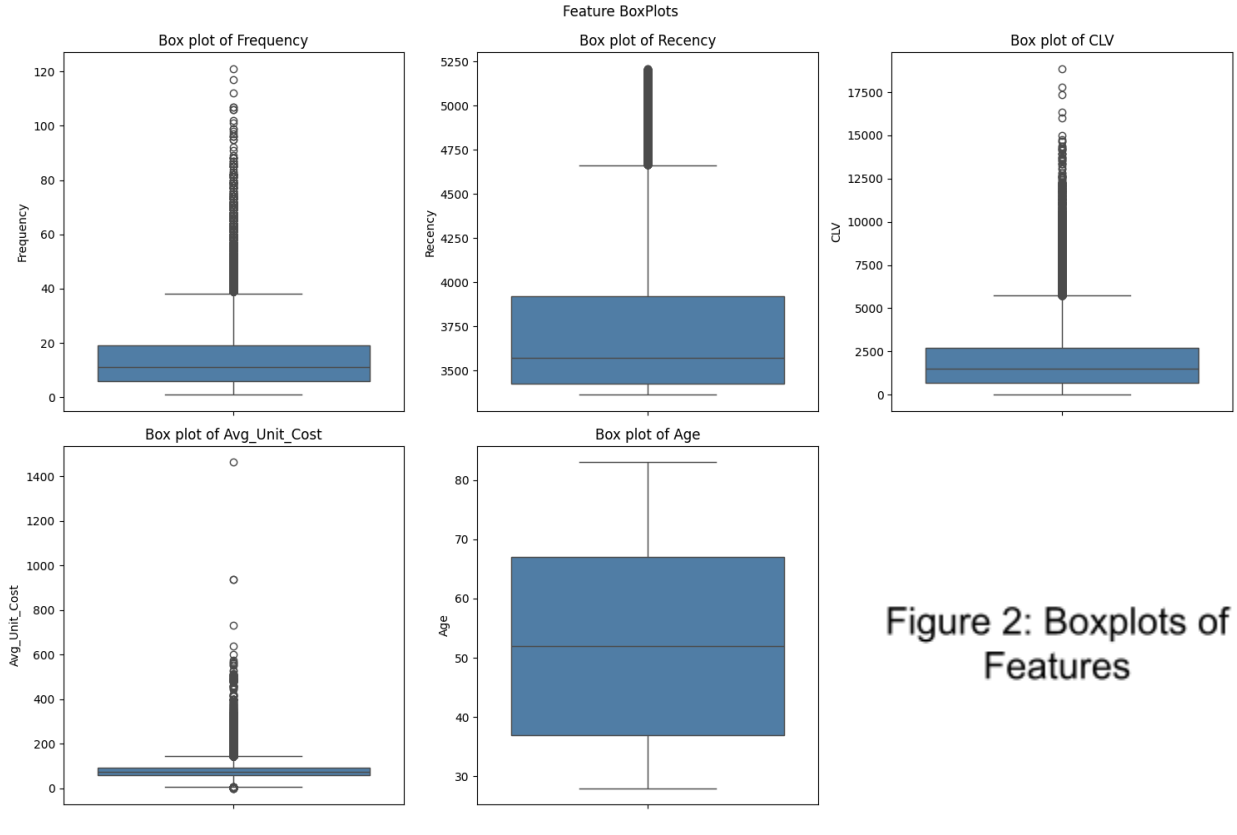

Figure 2: Boxplots of Features

Optimal Number of Clusters

Three complementary methods were used to determine the optimal k:

• Elbow Method (Figure 3) : Shows the rate of WCSS improvement slowing significantly from k=5 onwards, with drops beyond k=5 being roughly half those seen at lower values of k, supporting k=5 as the point of diminishing returns

• Silhouette Score: The score peaked at k = 2, indicating the tightest cluster cohesion. Next highest was at k=5. While k=2 statistically the strongest choice, it produces clusters that effectively split 66,000 customers into just two broad groups with limited business value.

• Hierarchical Clustering (Figure 4): The Ward linkage dendrogram (2,000-customer sample).The dendrogram does not strongly support k=5 in isolation, with the largest gaps suggesting k=2 and k=3. However, given the absence of statistically dominant clusters at any k, k=5 was selected on the basis of the elbow method, silhouette score, and business interpretability.

K=5 was selected at the optimal number of clusters.

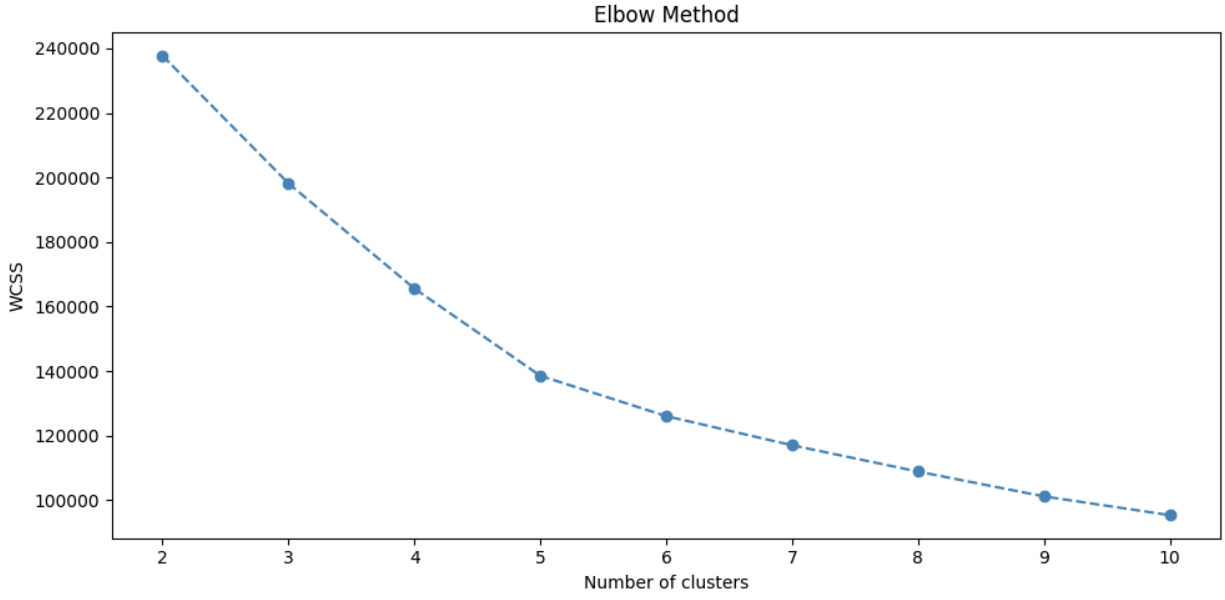

Figure 3: Elbow Method

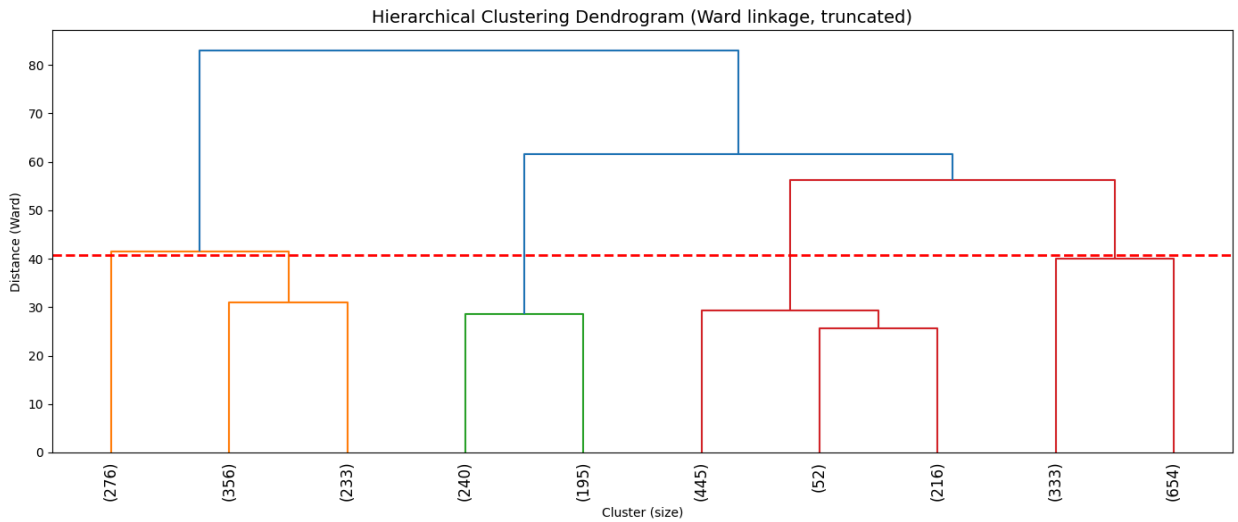

Figure 4: Dendogram

Results

Customer Segment Profiles — K-Means (k=5)

Cluster	Segment Name	Avg Frequency	Avg Recency (inverted)	Avg CLV	Avg Unit Cost	Avg Age
0	**Premium Buyers**	8.1	1,387	£1,783	£128.70	52
1	**Older Regulars**	10.7	1,598	£1,276	£65.60	69
2	**Lapsed Customers**	4.8	690	£550	£59.48	57
3	**High Value Active**	30.0	1,696	£4,387	£81.91	48
4	**Young Regulars**	12.2	1,594	£1,481	£67.53	37

** Recency (inverted): higher value = more recent purchase. CLV = Customer Lifetime Value (sum of Total Revenue per customer).*

The table above presents the mean feature values for each of the five customer segments identified through K-Means clustering (k=5) applied to 66,796 customers after outlier removal.

Cluster 3 (High Value Active) stands out as the most commercially significant segment, recording the highest mean frequency (30.0), recency (1,696) and CLV (£4,387), representing the most engaged and highest-spending customers. In contrast, Cluster 2 (Lapsed Customers)

records the lowest values across frequency (4.8), recency (690) and CLV (£550), identifying a disengaged segment requiring re-engagement. Cluster 0 (Premium Buyers) is distinguished by a markedly higher average unit cost (£128.70) relative to all other segments (£59–82), suggesting a niche group purchasing higher-value products at moderate frequency. Clusters 1 and 4 represent the two largest mid-value segments, differentiated primarily by age — older regulars (mean age 69) and younger regulars (mean age 37) — despite displaying similar frequency and CLV profiles. This data has also been visualised on box plots in Figure 5

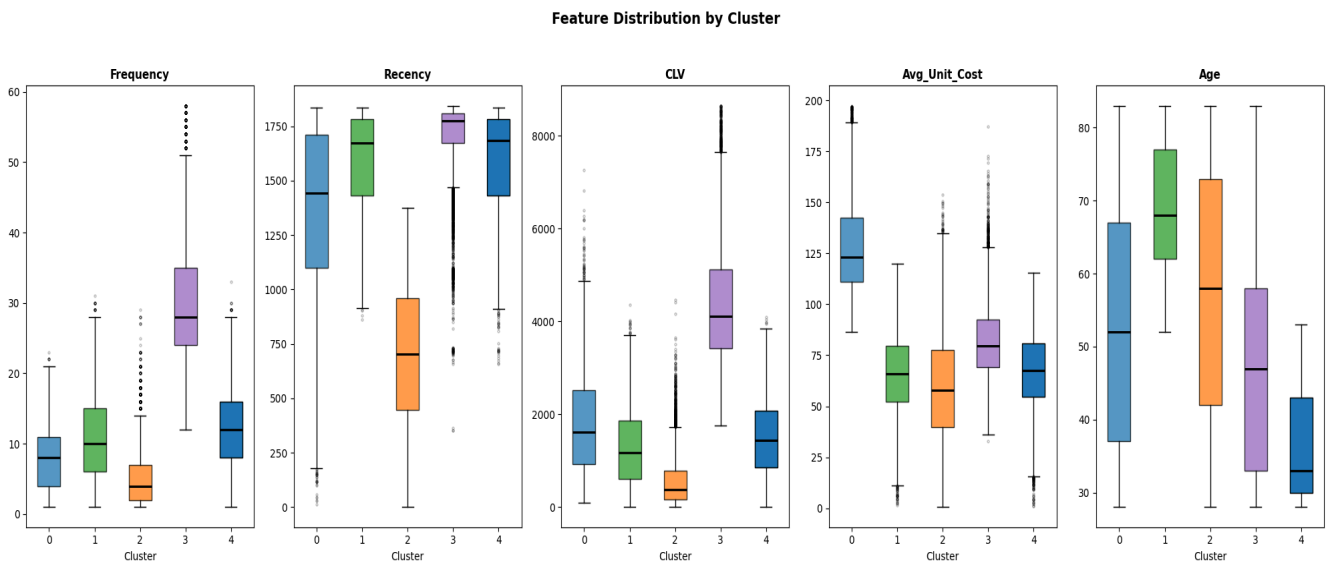

Figure 5: Cluster Boxplots

Dimensionality Reduction

PCA captured 65.6% of total variance in two components, providing a broad overview of cluster separation in Figure 6. Although the visual does not provide much for interpretation given the prevalent overlapping of clusters most likely caused by the 34.6% of variance that isn't captured in the 2D space.

t-SNE (applied to a 5,000-customer sample with perplexity = 40) seen in Figure 7 produced sharper, more compact cluster boundaries, confirming the five-segment structure with much less Overlap. Cluster 4 has the largest spread, and least tight boundary compared to all other clusters.

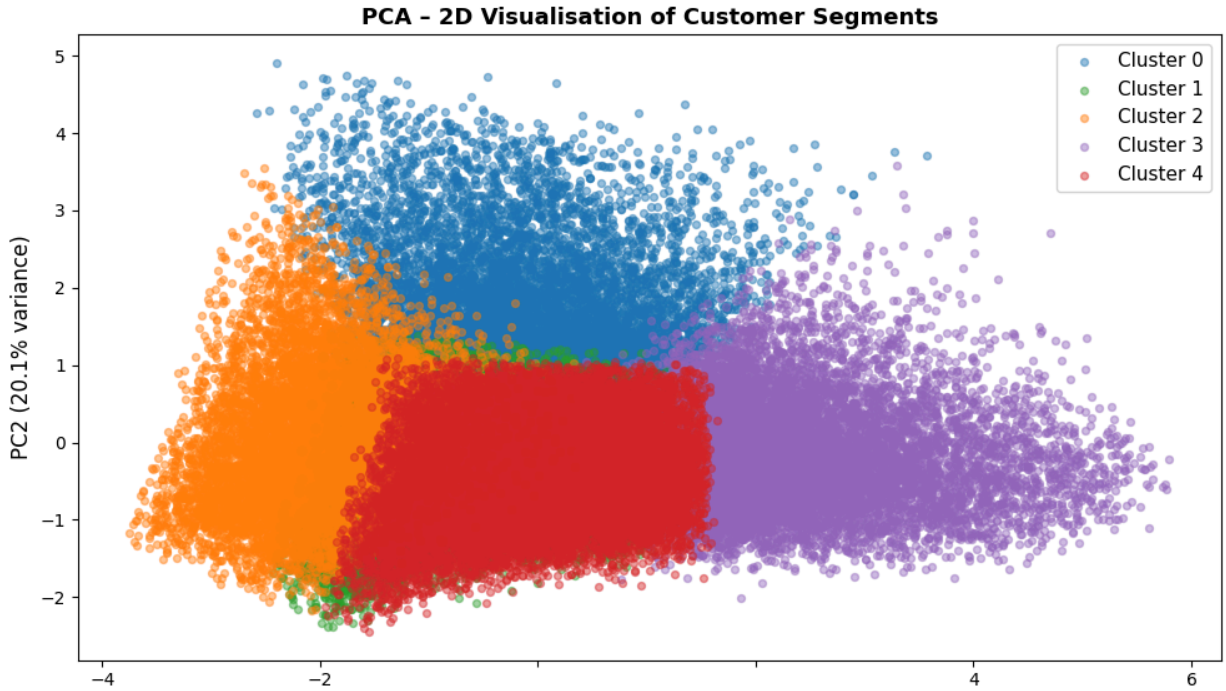

Figure 6: PCA Plot

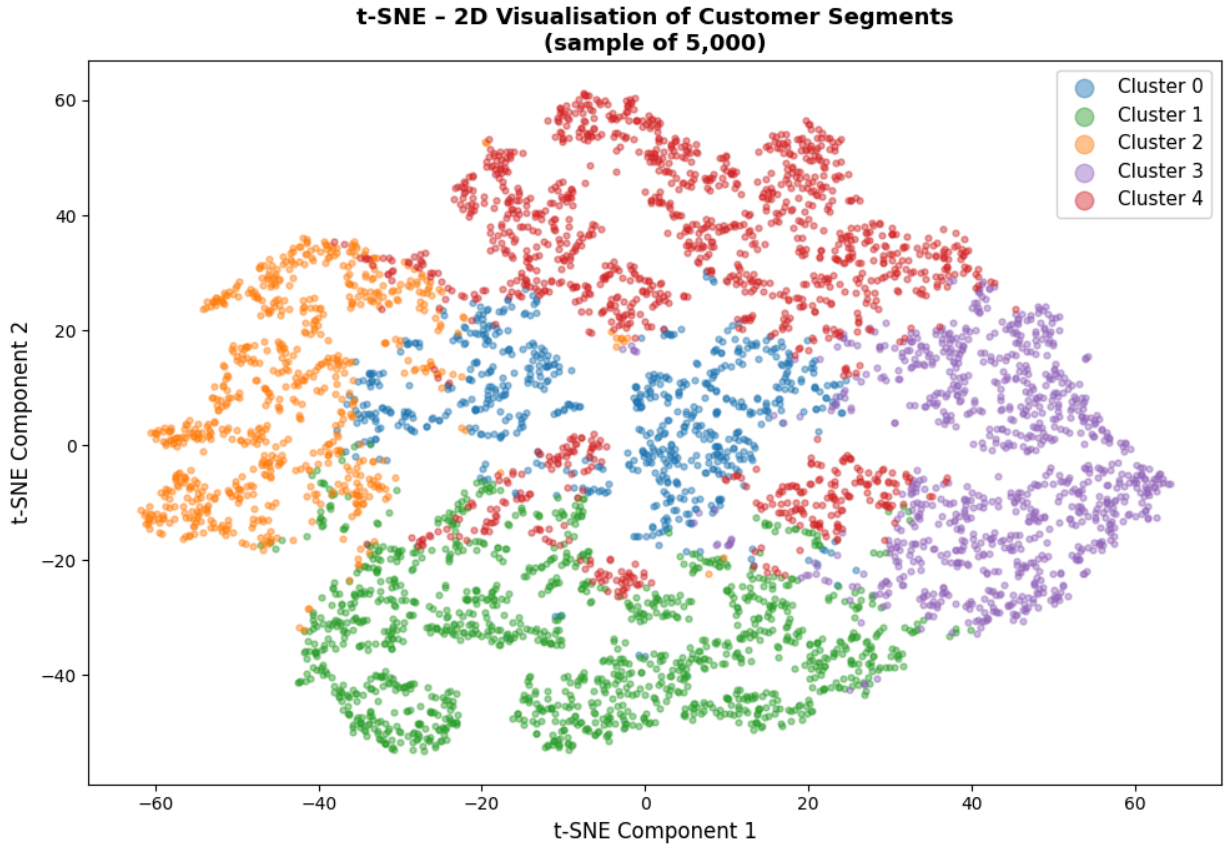

Figure 7: t-SNE Plot

Conclusion and Recommendations

Five meaningful customer segments were identified from 68,300 unique customers using K-Means clustering, validated by elbow, silhouette, and hierarchical methods. The following marketing strategies are recommended:

Cluster 3 — High Value Active: Prioritise retention through a dedicated loyalty programme, personalised communications, and exclusive product access. This segment contributes disproportionate revenue and losing customers here carries the greatest financial impact.

Cluster 2 — Lapsed Customers: Deploy a structured win-back campaign using time-limited discounts and personalised re-engagement emails referencing past purchases. Keep investment per customer controlled given their low CLV of £550.

Cluster 0 — Premium Buyers: Avoid volume-based promotions. Instead focus on curated high-value product recommendations and upselling strategies that reflect their willingness to spend significantly more per item (£128.70 average unit cost).

Clusters 1 and 4 — Older and Young Regulars: These two segments represent 54% of the customer base and the greatest growth opportunity. Despite similar CLV profiles, their contrasting ages (69 vs 37) suggest different channel preferences — traditional email and direct mail for Cluster 1, and digital-first engagement including social media and app-based offers for Cluster 4. Frequency-driving promotions such as bundle deals are recommended for both to grow CLV and migrate customers toward the High Value Active tier over time.

References

SAS. (2024). E-commerce customer dataset. Retrieved from the CAM DS101 course platform via FourthRev GitHub repository.